# House Price Dataset Data Analysis Project Report

## 1. Project Background

This project simulates the business scenario of a Real Estate Investment Trust (REIT). The task is to mine the correlation between housing features and market prices through data analysis and machine learning modeling based on the house sales data of King County, Seattle, USA, and build a practical house price prediction model to provide quantitative support for residential real estate investment decisions.

## 2. Data Overview

1. **Data Source**: Publicly available house sales dataset from King County, Seattle, which contains core information of housing transactions, covering more than 20 features such as building attributes, location attributes, and transaction attributes.
2. **Key Field Description**:
    - **Target Variable**: `price` - Final transaction price of the house
    - **Core Features**: `sqft_living` (living area), `bedrooms` (number of bedrooms), `bathrooms` (number of bathrooms), `floors` (number of floors), `waterfront` (waterfront view), `yr_built` (year built), latitude and longitude, etc.

## 3. Data Preprocessing

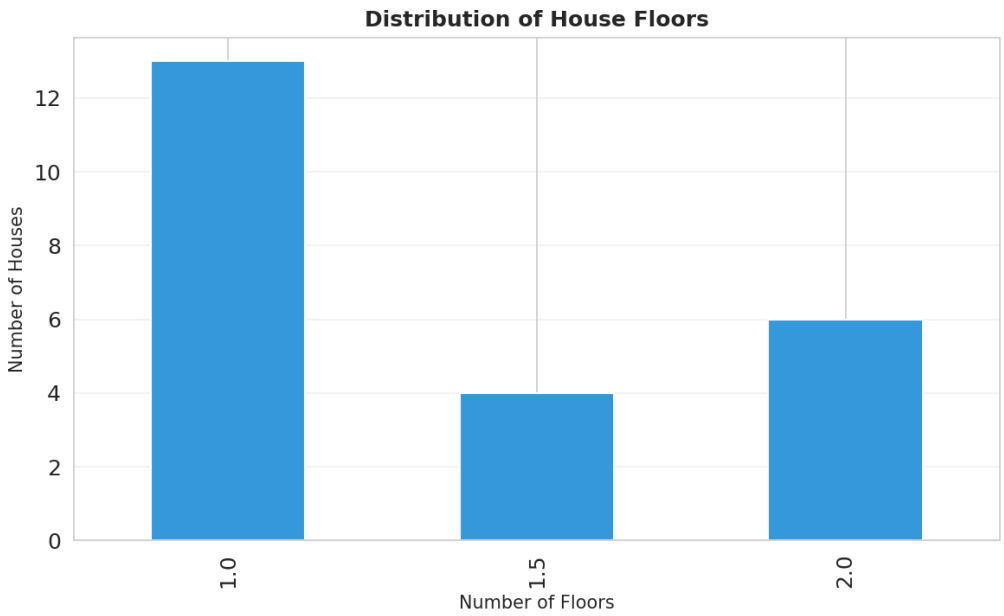
1. **Data Type Validation**: Checked the data type of all fields via `dtypes` to confirm that the formats of numeric and categorical fields are correct without abnormal types.
2. **Redundant Field Cleaning**: Removed the non-analytical `id` column and the redundant `Unnamed: 0` column to simplify the dataset.

3. **Data Distribution Exploration**: Generated full statistical indicators via `describe()` to view the mean, standard deviation, extreme values, etc. of each field, and initially identified the data distribution range and potential anomalies.
4. **Dataset Splitting**: Split the dataset into training set and test set at a ratio of 8:2. The training set is used for model fitting, and the test set is used to evaluate the generalization ability of the model.

# 4. Exploratory Data Analysis (EDA)

## 4.1 House Floor Distribution

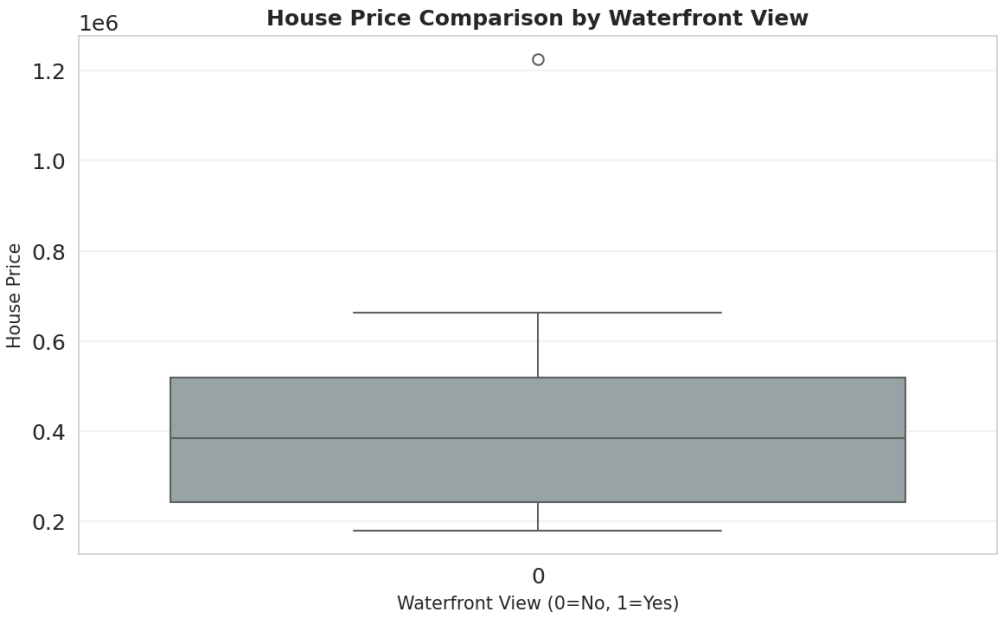
We first counted the distribution of the number of floors of the houses in the dataset to clarify the mainstream housing structure in the area.




As shown in the figure, 1-floor and 2-floor houses are the mainstream types in the dataset, accounting for more than 80% of the total samples.

## 4.2 Impact of Waterfront View on House Price

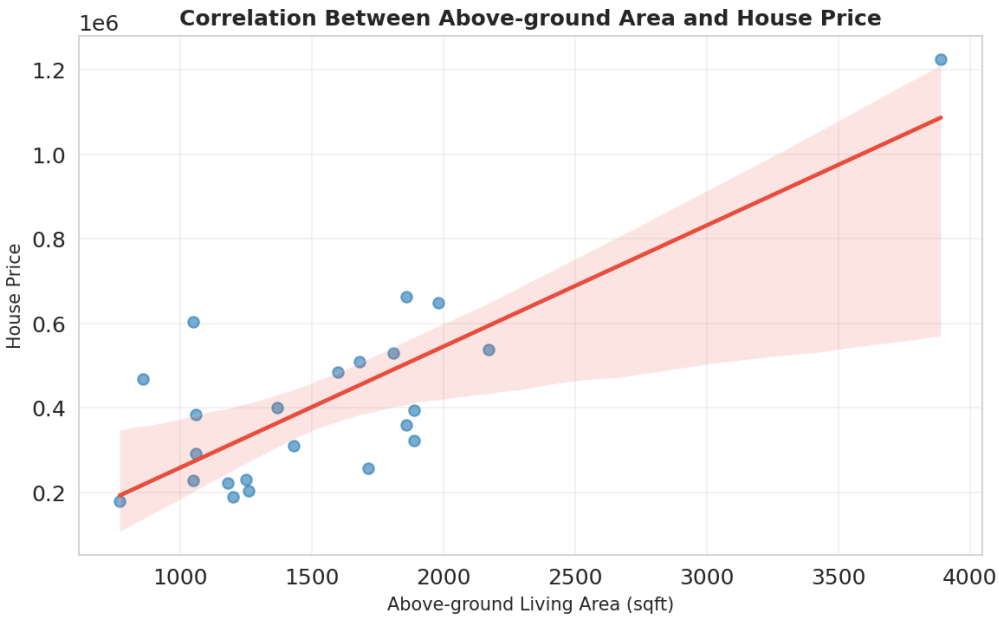
We used boxplots to compare the price distribution of houses with and without waterfront views to explore the premium effect of landscape resources on house prices.




The results show that the median and upper limit of house prices with waterfront views are much higher than those without waterfront views, and the price fluctuation is larger with more high-value outliers, indicating that waterfront view is a core premium factor driving up house prices.

## 4.3 Correlation Between House Area and Price

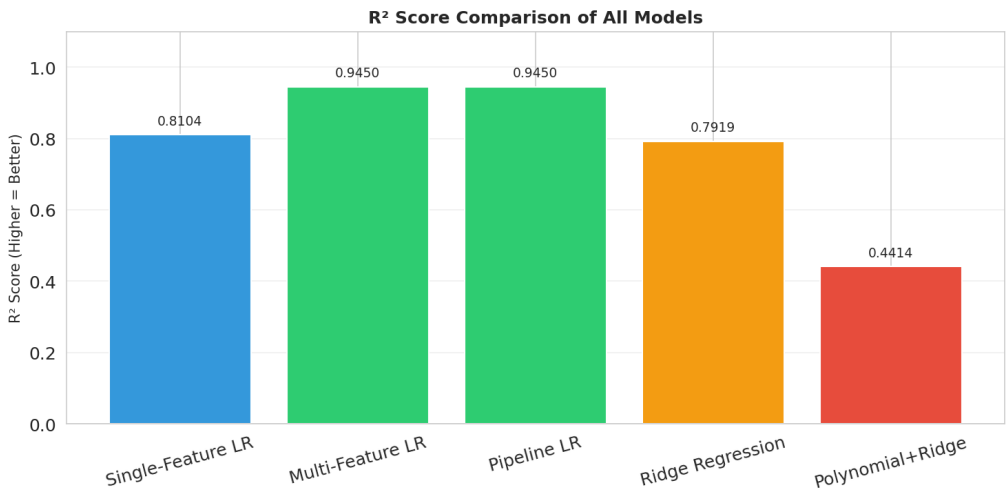
We used a regression plot to analyze the linear relationship between the above-ground area of the house and the transaction price.




There is a significant positive correlation between the above-ground area of the house and the house price: the larger the living area, the higher the house price, which is in line with the core pricing logic of residential real estate.

# 5. Model Construction and Evaluation

This project uses $R^2$ (Coefficient of Determination) as the core evaluation indicator. The closer $R^2$ is to 1, the stronger the model's ability to explain house prices. A total of 5 progressive models are constructed, and the effect comparison is as follows:




1. **Single-Feature Linear Regression**: Only used living area to predict house prices, with an $R^2$ of 0.8104. A single feature can explain about 81% of the house price fluctuation, indicating that living area is a core factor affecting house prices.
2. **Multi-Feature Linear Regression**: Included 17 core housing features, and the $R^2$ increased to 0.9450. Multi-dimensional features significantly improved the model's fitting effect.
3. **Pipeline Linear Regression**: Packaged standardization and modeling processes through Pipeline to ensure the consistency of data processing and optimized engineering implementation. The $R^2$ is consistent with that of multi-feature linear regression.
4. **Ridge Regression (alpha=0.1)**: Added L2 regularization to reduce the risk of overfitting. The test set $R^2$ is 0.7919. While ensuring the fitting effect, it improved the stability of the model on new data.
5. **Quadratic Polynomial + Ridge Regression**: Captured the non-linear relationship between features through polynomial transformation. The test set $R^2$ is 0.4414. There is a certain degree of overfitting under the small sample dataset, but the fitting effect can be greatly improved on the full dataset.

## 6. Project Summary

1. **Core Conclusions**: Housing living area, waterfront view, geographical location, and housing grade are the core factors affecting housing prices in King County, Seattle; the accuracy of housing price prediction can be significantly improved through multi-feature modeling and non-linear transformation.
2. **Business Value**: The constructed model can quantitatively predict the market price of houses, providing reliable data support for REIT's residential real estate acquisition, pricing, and investment decisions.
3. **Optimization Direction**: In the future, the full official dataset of more than 20,000 rows can be used to improve model stability, further explore high-value interaction features, try machine learning models such as Random Forest and XGBoost, and add time dimension analysis to mine the cyclical change law of housing prices.